

Load & Stress Test Report

Target: <https://mybook.lgacademy.com> · 22/05/2026, 19:09:04 (WIB) · Server: 4-core / 15 GiB VM · Tool: k6

As deployed today, the production server stays alive (zero failed requests) up to

~300 concurrent users

Recommended safe ceiling: 250 users — every request succeeds and 95% finish under 2 seconds. First failures appear at **325 users**; beyond **400** the server is overloaded. The limit is fixable on the same VM — see the action plan.

1 · CAPACITY RANGE — HOW MANY USERS IT HANDLES TODAY

Tested at 10 concurrency levels (identical profile: 15s ramp-up + 1 min sustained, realistic login + browse journey). The bar shows error rate.

CONCURRENT USERS	THROUGHPUT	ERRORS	LATENCY P95	ERROR RATE	STATE
100	136/s	0.00%	763 ms		Healthy
200	234/s	0.00%	1,726 ms		Healthy
250	286/s	0.00%	1,935 ms		Healthy
300	341/s	0.00%	2,081 ms		Alive · slow
325	345/s	1.33%	2,404 ms		Strained
350	363/s	4.13%	2,524 ms		Strained
400	390/s	7.44%	2,815 ms		Overloaded
600	337/s	7.99%	5,586 ms		Overloaded
800	359/s	10.54%	4,778 ms		Overloaded
1,000	399/s	12.14%	5,640 ms		Overloaded

Reading the range: Healthy up to 300 — every user served · Strained 325–350 — first failures (1–4%) · Overloaded 400+ — 7–15% of requests fail.

2 · THE PRODUCTION SERVER

COMPONENT	SPECIFICATION	NOTE
CPU	4 vCPU · Intel Xeon Silver 4214 @ 2.20 GHz	Only 1 core is used
Memory	15 GiB (10 GiB free)	Room for 3–4 replicas
Disk	55 GB (44 GB free)	Plenty of headroom
OS	Rocky Linux 9.7	Current

The server is not too small. It has 4 CPU cores; the app runs as one Node.js process using just 1 of them. Three cores sit idle even while the site fails — the fix is to use the hardware, not buy more.

3 · WHY IT BREAKS

- **One process ≈ one core.** The app caps at ~480 requests/second on a single core; 1000 users need ~1,500/s.
- **Static files overload it.** Each visit pulls 9 files; at scale that is thousands of parallel requests served by Node instead of a CDN.
- **Then connections queue and time out.** Once the queue is full, new requests are rejected — every failure is a timeout, not an app error.

4 · LLM ENDPOINT TEST (/API/V3/LLM)

⚠ Run while the AI Studio daily quota was exhausted — every call was quota-rejected upstream (free, no tokens billed). This measures the route handler under load, not real LLM speed. Re-test after the quota resets.

CONCURRENT USERS	REQUESTS	THROUGHPUT	RESPONDED	LATENCY P95	STATE
100	5,466	32/s	100.00%	1,144 ms	Healthy
1,000	17,741	65/s	96.93%	19,757 ms	Overloaded

The LLM route is much heavier than page serving — only ~65 req/s at 1000 users, with p95 collapsing to 19,757 ms and connections dropped. It also returns HTTP 500 for quota errors when it should return 429/503.

5 · WHAT TO DO — ACTION PLAN

Ordered by impact. **Items 1–3 alone clear 1000 users**, all on the existing VM with no hardware spend.

#	ACTION	IMPACT
1	Run 3 app replicas on the existing VM (it has 4 cores, uses 1) behind an nginx <code>upstream</code> block.	Very High
2	Add a CDN for static files (<code>/_next/static/*</code> , <code>/public/*</code>) — Cloudflare free tier.	Very High
3	Tune nginx: <code>worker_connections 4096</code> , <code>worker_rlimit_nofile 65536</code> , <code>upstream keepalive</code> .	Medium
4	Cache rendered pages — nginx micro-caching (1–10s) + <code>Cache-Control</code> headers.	High
5	Keep the VM (no upgrade needed). Add per-replica memory limits + a container healthcheck.	Medium
6	Fix LLM errors: return 429/503 instead of 500. Add monitoring + rate-limiting.	Medium
7	Future only — bigger VM / auto-scaling if you ever need well past ~2,500 users.	Low

6 · CAPACITY ROADMAP

STAGE	CORES USED	SAFE CONCURRENT USERS
Today — single process	1 of 4	~250–300
+ 3 app replicas (same VM)	4 of 4	~900–1,000
+ CDN for static assets	4 of 4	~1,500+
+ response caching	4 of 4	scales further

Every stage runs on the existing 4-core / 15 GiB VM. Re-run the k6 test after each change to confirm the new range.